

Ex. No 4: Analyze email headers and detect email spoofing using MHA (Mail Header Analyzer)

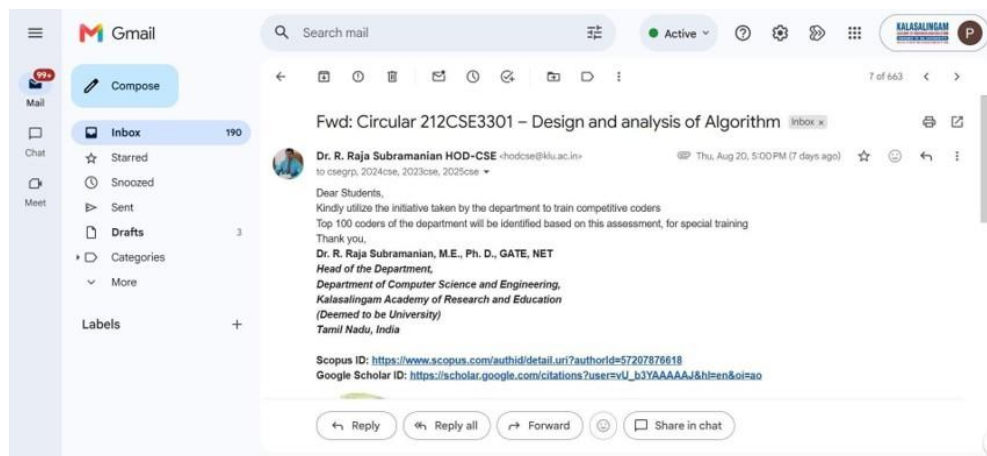
Date:

Description:

Here's a step-by-step guide on how to execute and analyze an email header:

1. Access the Email Header

- Gmail: Open the email, click the three dots (More) in the upper right corner, and select "Show original."
- Outlook: Open the email, click on "File," then "Properties," and find the "Internet headers" box.
- Yahoo: Open the email, click on the three dots (More), and select "View raw message."



2. Copy the Email Header

- Once you access the header, copy all the text that appears. This text contains various metadata fields about the email's journey from the sender to your inbox.

Original Message	
Message ID	<CAMq6mVb0q5dDE8nmG9csT7shp_YbLvZ+oYkwidsv7jCFJUKjA@mail.gmail.com>
Created at:	Thu, Aug 20, 2026 at 5:00 PM (Delivered after 19 seconds)
From:	"Dr. R. Raja Subramanian HOD-CSE" <hodcse@klu.ac.in>
To:	csegrp@klu.ac.in, 2024cse@klu.ac.in, 2023cse@klu.ac.in, 2025cse@klu.ac.in
Subject:	Fwd: Circular 212CSE3301 – Design and analysis of Algorithm
SPF:	PASS with IP 209.85.220.69 Learn more
DKIM:	'PASS' with domain klu.ac.in Learn more

3. Identify Key Header Fields

- From: The sender's email address.
- To: The recipient's email address.
- Date: The date and time the email was sent.

- Subject: The email's subject line.
- Return-Path: The return address if the email bounces.
- Received: A list of servers the email passed through on its way to the recipient. This is usually the most critical part of header analysis.
- Message-ID: A unique identifier for the email.
- SPF (Sender Policy Framework) / DKIM (DomainKeys Identified Mail): Indicators of email authentication.

4. Analyze the 'Received' Fields

- The Received fields show the path the email took from the sender to the recipient. These fields are listed in reverse order (from the last server to the first).
- Each Received line will show:
 - The sending server's hostname/IP address.
 - The receiving server's hostname/IP address.
 - The date and time the email passed through.

The screenshot shows the MXToolbox website's 'Email Header Analyzer' tool. The interface includes a navigation bar with links like 'SuperTool', 'MX Lookup', 'Blacklists', 'DMARC', 'Diagnostics', 'Email Health', 'DNS Lookup', 'Analyze Headers', and 'All Tools'. The main section is titled 'Email Header Analyzer' and contains a 'Paste Header:' text area with a sample email header. Below the text area is an 'Analyze Header' button and a 'Loading' indicator. At the bottom, there is an 'ABOUT EMAIL HEADERS' section with a brief explanation of the tool's purpose.

Sample Email Header:

```
--00000000000d66d77065978d8f6--
--00000000000d66d77065978d8f6
Content-Type: application/pdf; name="7UTF-87Q7212CSE3301_E2=80=93_Design_and_analysis_of_Algorithm=2Epdf?"
Content-Disposition: attachment; filename="7UTF-87Q7212CSE3301_E2=80=93_Design_and_analysis_of_Algorithm=2Epdf?"
Content-Transfer-Encoding: base64
Content-ID: <f_mt1fb640>
X-Attachment-Id: f_mt1fb640
--00000000000d66d77065978d8f6--
```

ABOUT EMAIL HEADERS

This tool will make email headers human readable by parsing them according to RFC 822. Email headers are present on every email you receive via the Internet and can provide valuable diagnostic information like

5. Check for IP Addresses and Hostnames

- Use tools like WHOIS or online IP lookup services to identify the geographical location and ownership of the IP addresses found in the Received lines.
- Check if any IP addresses are suspicious or if the hostname does not match the expected sending server.

6. Examine the SPF, DKIM, and DMARC Results

- SPF: Checks if the sender's IP is authorized to send emails for the domain.
 - Pass: Indicates the IP is authorized.
 - Fail: The IP is not authorized and could be a sign of spoofing.
- DKIM: Verifies that the email content has not been altered in transit.
 - Pass: The email has not been tampered with.

- SuperTool
MX Lookup
Blacklists
DMARC
Diagnostics
Email Health
DNS Lookup
Analyze Headers
All Tools

Header Analyzed

Email Subject: Fwd: Circular 212C5E3301 - Design and analysis of Algorithms

[Analyze New Header](#)

Delivery Information

 - DMARC Compliant
 - SPF Alignment
 - SPF Authenticated
 - DKIM Alignment
 - DKIM Authenticated

Relay Information

Received Delay:	6 seconds
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Relay Path	Delay (seconds)
From mail-sor141.google.com to mx.google.com	~1.2
to: tomx.google.com	~3.0
to: tomx.google.com	~4.0

- The Message-ID is unique to each email. It usually includes the domain of the sending server.

Hop	Delay	From	By	With	Time (UTC)	Blacklist
1	^	mail-sor41.google.com:250.08.220.41	mx.google.com	SMTPS	8/20/2026 11:30:29 AM	
2	2 seconds		2022a250402a00a6016a01933e134cc	SMTP	8/20/2026 11:30:31 AM	
3	3 seconds	mail-sor09.google.com:250.08.220.09	mx.google.com	SMTPS	8/20/2026 11:30:34 AM	
4	1 second		2022a2a83009c03039c704a9e0f	SMTP	8/20/2026 11:30:35 AM	

dmarc.klu.ac.in	Show	Solve Email Delivery Problems
<pre>v=DMARC1;pg=reporting;sp=none;ad=low;aspf=reject=100;fo=0;rfa=rf;ri=64480</pre>		
spr.klu.ac.in:208.85.220.69	Show	Solve Email Delivery Problems
<pre>v=spf1 ip4:208.115.225.170 include:entispancloud.com ~include:_spf_google.com ~all</pre>		
dkim.klu.ac.in:google	Show	
View Public Record <pre><v=DHnGj; b=sas; p=U3BzEj3d8gqCShwBAZCFACQ4BMTCgCQAQzFmYcyWmQJLds3XU1Sdd;BWVQ?y=FTHshQLUBR;rh=wkykx0tspenelx20r7xcKScCXlscFsmatCWFF/Sgujgd8L1P8q;v=N9NB;C5ztp;ad=jzll;eL1MCEFB?Qzfpw=C38dDC7C2dCS5wVrVSstymya(5j67necSwz)/cmre;ig=pfIX+Llmg=hdvcrbaizgvCvClr3jgmff7hsTgnfCL7Lkshsws0d05d4<</pre> View Signature <pre>v=3; a=s-as-dns256; c=rrelaxed/relaxed; d=klu.ac.in; i=_google; t=1371225434; u=137338134; dm=klu.ac.in; h=list-unsubscribe-list-archive-list-help-list-post-list-id-mailing-list-precedence-s-original-authentication-results-ia-original-sender-content-type-to-subject-message-id-date-from-list-reply-to-references-name-version:from:</pre>		

- **Mismatch in Email Domains:** Check if the From domain matches the domains in ReturnPath, Message-ID, and other authentication results.
- **Strange IP Addresses:** Be wary of IP addresses that do not match the known sending server or are from unexpected locations.

- Check IP Addresses: Lookup 192.0.2.1 and ensure it is associated with mail.example.com.
- Verify SPF/DKIM: Check if mail.example.com is authorized to send emails for this domain.

- Compare Time Stamps: Ensure the times make sense chronologically without unusual delays.

Rubrics:

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

RESULT

The email header was successfully extracted and analyzed using MXToolbox. Authentication protocols (SPF, DKIM, DMARC) and the originating IP address were manually verified, demonstrating how to trace email origins and detect potential spoofing